



Module 4 Lecture - States of Consciousness

Introductory Psychology

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Understand what is meant by consciousness
- Explain how circadian rhythms are involved in regulating the sleep-wake cycle, and how circadian cycles can be disrupted
- Discuss the concept of sleep debt
- Describe areas of the brain involved in sleep
- Understand hormone secretions associated with sleep
- Describe several theories aimed at explaining the function of sleep
- Name and describe three theories about why we dream
- Differentiate between REM and non-REM sleep
- Describe the differences between the three stages of non-REM sleep
- Understand the role that REM and non-REM sleep play in learning and memory
- Describe the symptoms and treatments of insomnia
- Recognize the symptoms of several parasomnias
- Describe the symptoms and treatments for sleep apnea
- Recognize risk factors associated with sudden infant death syndrome (SIDS) and steps to prevent it
- Describe the symptoms and treatments for narcolepsy
- Describe the diagnostic criteria for substance use disorders
- Identify the neurotransmitter systems impacted by various categories of drugs
- Describe how different categories of drugs affect behavior and experience
- Define hypnosis and meditation
- Understand the similarities and differences of hypnosis and meditation

1.2 Instructor Learning Objectives

- Understand how consciousness is better described as range of states, rather than simply “on” or “off”
- Know several common conditions and drugs that modify or otherwise change the state of consciousness

1.3 Introduction

- You ever been in the back of a classroom, watching a boring lecture, slowly feeling yourself falling asleep?
 - We’ve all been there...
 - All of us have had the _____ of being wide awake, being tired/-fatigued, being asleep, etc.
 - We’ve also likely

! Important

How many hours of sleep do you get each night, and do you regularly feel awake or tired?

- There are also other states of consciousness, related to use of specific substances or _____ disassociation

2 What is Consciousness?

2.1 Introduction

! Important

There is a more philosophical debate to have on what exactly 'consciousness' is - we are going to try to stay more focused on a factual concrete description today, but it is worth reading into more!

- **Consciousness** is our state of _____ of our environment, perceptions, and thoughts.
- When we are _____, we are usually at least reasonably conscious of our surroundings, but when we are **sleeping** we are more-or-less completely unaware
 - But between these two states, we have things like _____, day-dreaming, being intoxicated with substances, etc.

2.2 Biological Rhythms

- Our bodies _____ on many rhythmic biological processes:
 - Sleep and awake
 - Body temperature natural fluctuations
 - Menstrual cycles
 - Many others
- Our bodies attempt to maintain **homeostasis**, which is a restful, optimal state in which _____, cells, substances, and other bits of our body are in balance
 - All of these _____ help this process of homeostasis
- Our brain has built-in _____ called the **suprachiasmatic nucleus** in the hypothalamus

- However, one nuance of this structure is that it is connected to the detection of light (usually sunlight). Constant light or lack thereof could possible disrupt proper function.

2.3 Problems with Circadian Rhythms

2.4 Disruptions of Normal Sleep

2.5 Insufficient Sleep

3 Sleep and Why We Sleep

3.1 Introduction

3.2 What is Sleep?

3.3 Why Do We Sleep?

3.4 Adaptive Function of Sleep

3.5 Cognitive Function of Sleep

4 Stages of Sleep

4.1 Introduction

4.2 NREM Stages of Sleep

4.3 REM Sleep

4.4 Dreams

5 Sleep Problems and Disorders

5.1 Introduction

5.2 Insomnia

5.3 Parasomnias

5.4 Sleepwalking

"I don't mind not knowing. It doesn't scare me." — Richard P. Feynman

5.5 REM Sleep Behavior Disorder (RBD)